

CoRoT-1b

R-2605

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_151115 · created 2026-08-28T01:03:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457341.8629 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.041
Transit depth (Rp/Rs)^2	1.98 +/- 1.16 [%]
Orbital Inclination (inc)	88.54 +/- 2.85
Airmass coefficient 1 (a1)	0.54 +/- 0.37
Airmass coefficient 2 (a2)	0.36 +/- 0.71
Scatter in the residuals of the lightcurve fit is	61.31 %
Best Comparison Star	#2 - [301, 399]
Optimal Aperture	1.34
Optimal Annulus	6.25
Transit Duration (day)	0.1112 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.041 vs 0.1388 (deviation 0.05 σ)
Duration fit vs published	0.1112 d vs 0.1046 d (deviation 0.9 σ)
Mid-time O-C	-7.3 min
Depth SNR	3.1
Residual scatter	61.31 %

Light curve

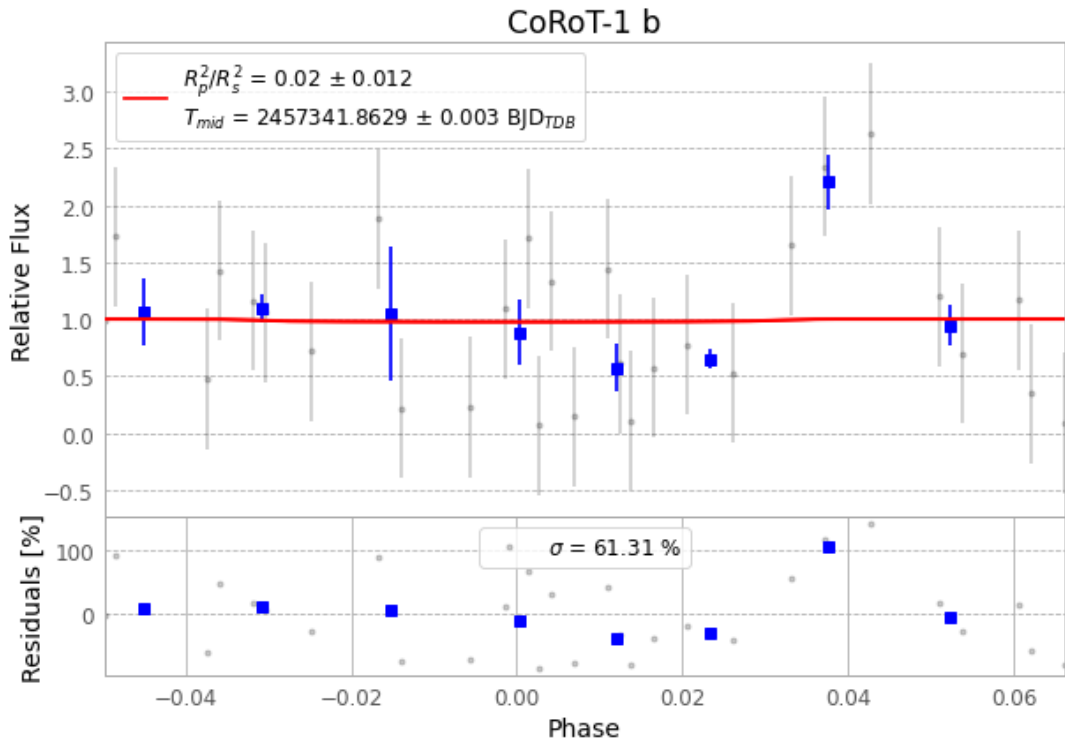


Figure 1 — FinalLightCurve_CoRoT-1 b_2015-11-14.png (SHA-256 bd992c92c58113a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [239, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 873ae12f679e10bc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

84 FITS frames (55 MB), CoRoT-1151115064512.FITS → CoRoT-1151115110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-151115.log (123e8f8e4d3d...), FinalLightCurve_CoRoT-1 b_2015-11-14.png (bd992c92c581...), FinalLightCurve_CoRoT-1 b_2015-11-14.csv (51be23e5a1bf...)

Compute resources

Wall time 133.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:03Z.